University of Illinois at Chicago
Department of Chemical Engineering

SAFETY QUIZ: CHE381- Chemical Engineering Laboratory I

Please answer the following questions testing your knowledge of the content of the Chemical Hygiene Plan safety document. Be brief but concise in your answers. Open book/open notes and team discussion is allowed but answers must be in your own words.

1. What is the most fundamental rule of chemical safety?

Awareness is the single most fundamental rule of chemical safety

2. What two pieces of personal protective equipment must be worn at all times in the lab?

Safety glasses and safety shoes

3. What additional kinds of personal protective equipment are available depending on the chemical hazard or experimental conditions?

Lab coats, face shields, gloves, splash aprons, ear protection

4. True or False: Fume hoods are generally used as storage areas for chemicals.

False, they are used for conducting experiments and temporary storage of chemicals.

5. Fume hoods are to be used when the Threshold Limit Value (TLV) of a chemical is at or below:

TLV of less than 1000 ppm

6. What are the four routes of chemical entry into the body?

Inhalation, ingestion, injection, eye/skin contact

7. True or False: Respirators are generally used in the R&D laboratories for normal work procedures.

False, respirators are generally not used for normal procedure, periodic use is OK

8. What is the role of the Chemical Hygiene Plan?

This Chemical Hygiene Plan describes American Air Liquide's safety program, including (but not limited to) chemical safety rules, personal protective equipment, control equipment (i.e., fume hoods), employee training programs, and medical programs. The Chemical Hygiene Plan is designed as a tool to coordinate safety procedures.

9. What is the guideline for laboratory room airflow and fume hood airflow?

Laboratory airflow: 20 linear feet per minute, Fume hood: 80-120 linear feet per minute

10. List 3 Common Housekeeping Practices that promote chemical hygiene and safety:

Full List (pick 3):

- **Keep all aisles, hallways, and stairs clear of all chemicals.**
- **Keep all work areas and especially workbenches clear of clutter and obstructions.**
- **All working surfaces and floors should be cleaned regularly.**
- **Access to emergency equipment, showers, eyewashes and exits should never be blocked by anything.**
- **Wastes should be properly labeled, and kept in the proper containers.**
- **Any unlabeled containers are considered wastes by the end of each workday.**

11. What is the proper method of transporting gas cylinders in the laboratory?

Gas cylinders must be transported using a 1-cylinder 4-wheel cart.

12. True or False: MSDS sheets are required only for toxic chemicals and gases used in the laboratory.

False, MSDS sheets are required for all chemicals and gases used in the laboratory.

13. What is an accepted method of vapor detection in the laboratory?

Accepted methods of vapor detection; safety monitors, detector tubes (Draeger tubes), detection meters. Odor detection is not an accepted method.

14. Name 3 Personal Hygiene Practices that should be followed in the laboratory.

Pick 3:

- **Wash promptly if skin contact is made with any chemical, regardless of corrosivity.**
- **Wear appropriate eye protection at all times.**
- **Avoid inhalation of chemicals, do not "sniff" test chemicals.**
- **Do not mouth pipette anything, use suction bulbs.**
- **Wash well before leaving the laboratory; do not wash with solvents, use soap.**
- **Change clothing as soon as possible after leaving laboratory, and launder work clothes often.**
- **Do not eat or drink in chemical areas.**
- **Do not bring food, beverage or tobacco products into chemical storage or use areas.**

15. Describe how individual chemicals of less than 15 lbs. (for example a bottle of acetone) are safely transported.

Individual chemicals in a container are transported using rubber “bucket-type” carriers.

16. List the Emergency Procedure to be followed for a large chemical spill or toxic gas release in the laboratory.

- **Evacuate the immediate area.**
- **Dial "0" (Facility Operator), advise nature of emergency, and location of emergency (e.g., "Chemical Spill, XYZ lab").**
- **The Operator will notify all employees (on the Public Address system) to evacuate, and will activate the Emergency Response Team.**
- **Immediately exit the lab and deactivate the lab power by pushing the emergency (red) power button outside of the lab.**
- **Evacuate the building through the nearest and safest exit.**
- **Unless otherwise notified (on the P.A. system), assemble in front of the facility at the middle driveway.**
- **Report to the department secretary**

17. What is the difference between a flammable chemical and a combustible chemical in terms of OSHA and NFPA guidelines.

"Flammable" is generally used to refer to chemicals with a flash point below 100° F. Chemicals with flash points between 100 and 200° F are termed "Combustible".

18. List the Emergency Procedure for an individual chemical exposure in the laboratory.

In the case of a chemical exposure (of a worker), the following steps should be observed:

- **Dial "0" (Facility Operator), advise nature of emergency, and location of emergency**
- **(E.g., "Person with severe chemical burns, ABC lab").**
- **Operator calls Fire Department Paramedics and notifies Emergency Response Team.**
- **If there are no apparent dangers to yourself of chemical exposure, the following may be performed until the E.R.T./Paramedics arrive:**
- **If a person has been splashed with a chemical, wash them with plenty of water for at least 15 minutes, removing all contaminated clothing (Eyewashes for eye/face splashes, Emergency Shower for all other).**
- **If a person has been overexposed by inhalation, get victim to fresh air (if it is safe for you to enter area), apply artificial respiration if necessary.**

19. What is the definition of a “hazardous chemical”?

A chemical for which there is statistically significant evidence that an acute or chronic health effect may occur in exposed employees.

20. All laboratories that handle chemicals must be equipped with eyewashes and safety showers. What is the recommended flowrate of this equipment and how often should they be checked?

Safety shower: 30 gallons per minute. Eye showers: 3 gallons per minute. Both checked monthly.